

Author Disclosure: R.L. Deraniyagala: None. C.M. Bryant: None. C.G. Morris: None. W.M. Mendenhall: None. D.J. Indelicato: None. J.A. Bradley: None. R. Dagan: None.

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Clinical Trial of Induction Chemotherapy and Concurrent Chrono-Chemotherapy With Radiation Therapy in Locally Advanced Nasopharyngeal Carcinoma

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Purpose/Objective(s): To compare the Short-term efficacy and observe the safety of chemotherapy plus concurrent chrono-chemotherapy followed by intensity modulated radiation therapy in locally advanced nasopharyngeal carcinoma.

Materials/Methods: From December 2013 to November 2014, we enrolled 96 patients with stage III-IVb nasopharyngeal carcinoma from Guizhou Cancer Hospital of China. Both study and control group received 2 cycles induction chemotherapy (Docetaxel 75mg/m², infusion, d1. Cisplatin 75mg/m², infusion, d1. 5-fluorouracil 750mg/m²/d, continuous intravenous pumping, d1-5(120 hours), 21 days/cycle, for two cycles), and 3 cycles of concurrent chrono-chemotherapy with IMRT were gone through in the study group (cisplatin, 100 mg/m², 10:00-22:00, continuous intravenous pumping, d1, 21 days/cycle, for three cycles) while the control group received 3 cycles of conventional chemotherapy with IMRT. The two groups were compared in terms of recent curative effect, overall survival (OS) and adverse effects.

Results: With a median follow-up of 19 months, to compare with the control group, the study group has a better outcome in recent adverse effects and immune functions. During the phase of concurrent chemoradiation therapy, the incidence of nausea, vomiting and oral mucositis in two groups was 72.9%: 81.2% ($P < 0.05$), 52.1%:79.2% ($P < 0.05$), 77.9%:89.6% ($P < 0.05$) respectively. After concurrent chemoradiation therapy, CD4+/CD8+ in study group was significantly in better results (1.7±0.9 against 1.2±0.8, $P < 0.05$). There were no significant differences in curative effect and overall survival between the two groups. The complete remission rate of nasopharyngeal tumor (97.9%:95.7%, $P > 0.05$). The 2-year OS for the study group were 86.5% versus 87.7% for the control group ($P > 0.05$). The trial is registered at ClinicalTrials.gov, number NCT02187315.

Conclusion: Induction chemotherapy plus concurrent chrono-chemotherapy followed by intensity modulated radiation therapy may become a proper treatment for its remaining curative effect while reducing adverse effects and improving immune function at the same time. Further research is required for collecting more samples to obtain further objective appraisal.

Author Disclosure: X. Zhang: None.

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Clinical and Volumetric Outcomes of Gated Proton Beam Therapy for Choroidal Melanoma in Korea

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Purpose/Objective(s): This study aimed to assess the clinical benefits of automatic beam gating system in the proton therapy for intraocular uveal melanoma

Materials/Methods: Total 20 patients who underwent proton beam therapy for uveal melanoma between 2009 and 2014 were reviewed. In-house automatic beam gating system was used to prevent a misalignment caused by eye movement during the treatment. The median basal tumor diameter was 11 mm (range 5-16 mm), and the median tumor thickness was 8 mm (range 2-13 mm). Dose prescriptions were 60 PGE (Proton Gray Equivalent)/5fx (7 patients), 70 PGE/5fx (12 patients) and 71 PGE/5fx (1 patient).

Results: The median follow-up duration was 43 months (Range 10-76 months). The local control rate was 95%. In addition, disease control rate

was 90%. Acute high grade complications were vitreous hemorrhage and cataract. Late high grade complications were posterior synechia and neovascular glaucoma. Four patients maintained their visual acuity (VA). Patients of 70% had less than 0.1 VA. By the results of volumetric evaluation, the mean volume of tumor was 0.6294 mL, 0.5078 mL, 0.41 mL and 0.39 mL at baseline, 1st follow-up, 2nd follow-up and 3rd follow-up. Of 20 patients, one patient was performed enucleation. Only one patient had increased tumor volume and others decreased tumor volume.

Conclusion: Use of automatic beam gating system in proton eye treatment may be more effective on tumor control, tumor reduction and prevention of ocular complications compared to that of non-gated therapy.

Abstract 2913; Table 1.

Patient	Baseline	1 st F/U	2 nd F/U	3 rd F/U
1	1.48	1.02	Enucleation	
2	0.418	0.403	0.389	0.354
3	0.733	0.779	0.704	0.875
4	0.088	0.067	0.057	0.049
5	0.405	0.391	0.35	0.321
6	0.56	0.517	0.499	0.434
7	0.313	0.383	0.315	0.228
8	0.245	0.15	0.146	0.11
9	0.567	0.424	0.368	0.323
10	0.653	0.576	0.411	0.373
11	0.755	0.567	0.375	0.344
12	1.61	1.37	1.13	0.959
13	0.744	0.634	0.54	0.464
14	0.095	0.084	0.066	0.052
15	0.909	0.714	0.637	0.579
16	0.761	0.632	0.577	0.414
17	1.04	0.87	0.739	0.602
18	0.825	0.352	0.307	0.194
19	0.255	0.128	0.089 (Last)	
20	0.133	0.094	0.062 (Last)	
Mean	0.629±0.419	0.508±0.475	0.410±0.277	0.390±0.254

Author Disclosure: E. Choi: None. J. Park: None. D. Shin: None. J. Kim: None. J. Kim: None. S. Moon: None.

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Oral Cavity Cancer Surgical Pathologic Factors Influence Accuracy of 3-Month Postradiation Therapy FDG Positron Emission Tomography/Computed Tomography

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Purpose/Objective(s): Recurrence monitoring for oral cavity squamous cell carcinoma (SCC) following definitive surgery and adjuvant (chemo) radiation therapy (RT) with F-18 Fluorodeoxyglucose (FDG) PET/CT is not well established. This study evaluates the accuracy of 3 month post-RT PET/CT in oral cavity cancer patients treated with surgery and adjuvant (chemo)RT, with an additional objective evaluating patient tumor characteristics which influence PET/CT accuracy.

Materials/Methods: Oral cavity SCC patients receiving definitive surgical resection, adjuvant (chemo)RT, and 3 month post-RT PET/CT were retrospectively examined between 2005 and 2015. PET/CTs were classified as positive or negative using the nuclear medicine interpretation, with equivocal scans assigned positive if the ordering clinician performed biopsy, planned for closer follow-up, or ordered additional imaging. True or false results were based on clinical follow-up within one year of PET/CT. Imaging accuracy was calculated separately for locoregional and distant disease. Further analysis evaluated associations between 3 month post-RT PET/CT accuracy, and disease sub-site and pathologic risk factors including positive lymph nodes, extracapsular extension, perineural invasion (PNI), lymphovascular invasion, bone invasion, positive margin, and pathologic grade.

Results: Fifty-six patients met inclusion criteria; with 14 (25%) receiving concurrent chemoRT. Median time from RT end to PET/CT was 101 days. Twenty-three (41%) had a positive scheduled surveillance 3 month-post RT PET/CT: 14 (25%) had PET/CTs that were positive in the head and